Metabolite ID MP000351

Berberine

KEGG PubChem KNApSAcK COCONUT

SECTIONS

Chemical Information
Natural Product Classification
Pathway Information
Potential Producer Species
Reported source species

Chemical Information

Metabolite ID	MP000351	**Name**	Berberine
Molecular formula	C20H18NO4	**Molecular weight**	336.36
PubChem CID	2353	**KEGG ID**	C00757
CAS number	2086-83-1	**IUPAC name**	16,17-dimethoxy-5,7 azoniapentacyclo[11 1(13),2,4(8),9,14,16
SMILES	COC1=C(C2=C[N+]3=C(C=C2C=C1)C4=CC5=C(C=C4CC3)OCO5)OC	**Synonyms**	Berberine;Umbellati

Natural Product Classification

SUPER CLASS **Alkaloids**	NP PATHWAY **Alkaloids**
NP SUPERCLASS **Tyrosine alkaloids**	NP CLASS **Isoquinoline alkaloids|Protoberberine alka**

Pathway Information

Pathway records indicate whether this metabolite appears as a precursor, intermediate, or end product.

Pathway	Role	
Berberine Biosynthesis Pathway	End product	

Potential Producer Species

4 species reach 100% route completeness in at least one pathway.

Taxa distribution

Expand taxonomic groups to reach species-level completeness scores.

Search Latin or Chinese name

Viridiplantae 3

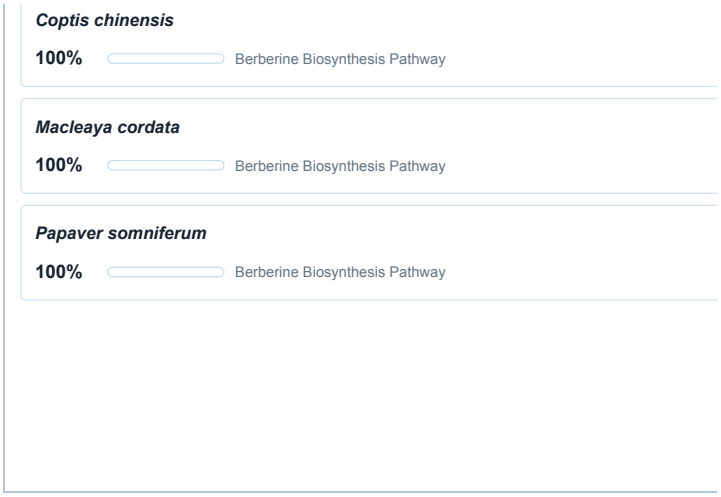
All producer species

4 producer species in the selected taxon.

Cinnamomum kanehirae

100% Berberine Biosynthesis Pathway





Reported Source Species

Species reported as sources of this metabolite in public databases.

Metabolite	Species	Herb	Knapsack
Berberine	*Coptis chinensis*	-	-
Berberine	*Dunaliella salina*	-	-
Berberine	*Macleaya cordata*	-	-
Berberine	*Magnolia biondii*	-	-
Berberine	*Papaver somniferum*	-	-
Berberine	*Ziziphus jujuba*	-	-

NGDC

National Genomics Data Center

86-10-84097298

ngdc(at)big.ac.cn

Policies and Disclaimers

This database is free for academic use only. For any commercial use, please contact us for commercial licensing terms.

Research & Resources
Databases
Tools
Standards
Topics & Projects
Publications
Annual Reports

Featured
Database Commons
GenBase
GSA
GWH
GVM
GEN
RCoV19

Alliance & Collaboration
BHBD
P10K
Partners
Collaborations
Funding
Contact Us
Join Us

Conference & Outreach
Conferences
Education

About
About NGDC
People
Mission
Advisory Board
Organizational Structure
History

国家生物信息中心
CHINA NATIONAL CENTER FOR BIOINFORMATION

© 2026 China National Center for Bioinformation / Beijing Institute of Genomics , Chinese Academy of Sciences
No.1 Beichen West Road, Chaoyang District, Beijing 100101, China
京ICP备10050270号